
Conditional neural control variates for variance reduction in Bayesian inverse problems

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Abstract

Bayesian inference for inverse problems involves computing expectations under posterior distributions—e.g., posterior means, variances, or predictive quantities—typically via Monte Carlo (MC) estimation. When the quantity of interest varies significantly under the posterior, accurate estimates demand many samples—a cost often prohibitive for partial differential equation-constrained problems. To address this challenge, we introduce conditional neural control variates, a modular method that learns amortized control variates from joint model–data samples to reduce the variance of MC estimators. To scale to high-dimensional problems, we leverage Stein’s identity to design an architecture based on an ensemble of hierarchical coupling layers with tractable Jacobian trace computation. Training requires: (i) samples from the joint distribution of unknown parameters and observed data; and (ii) the posterior score function, which can be computed from physics-based likelihood evaluations, neural operator surrogates, or learned generative models such as conditional normalizing flows. Once trained, the control variates generalize across observations without retraining. We validate our approach on stylized and partial differential equation-constrained Darcy flow inverse problems, outperforming classical Stein control variates and achieving substantial variance reduction, even when the analytical score is replaced by a learned surrogate.

1 INTRODUCTION

Quantifying uncertainty in the solution of inverse problems is of fundamental importance in many areas of science and engineering [Tarantola, 2005, Stuart, 2010]. Due

to noise and ill-posedness, a single point estimate fails to capture the non-uniqueness of the solution. The Bayesian framework addresses this by characterizing the full posterior distribution over solutions consistent with the observed data and prior knowledge. Posterior samples then provide expectations of quantities of interest—e.g., posterior means, variances, and credible intervals. Computing these expectations requires evaluating integrals of the form $\mathbb{E}_{p_{\text{post}}(\mathbf{x}|\mathbf{y})}[h(\mathbf{x})]$, which are typically intractable in closed form. While Monte Carlo (MC) estimation replaces the integral with a sample average [Robert and Casella, 2004], its variance— $\text{Var}(h(\mathbf{x}))/N$ for N i.i.d. samples—can be prohibitively large when $h(\mathbf{x})$ varies significantly under the posterior. In partial differential equation (PDE)-constrained inverse problems, where each posterior sample requires a computationally expensive forward solve, this high variance translates into high computational cost.

Control variates offer a classical remedy [Fieller and Hartley, 1954, Robert and Casella, 2004]: given an auxiliary function $g(\mathbf{x})$ with zero expectation under the target distribution and positive correlation with $h(\mathbf{x})$, the controlled estimator $\mathbb{E}_{p_{\text{post}}(\mathbf{x}|\mathbf{y})}[h(\mathbf{x}) - g(\mathbf{x})]$ has lower variance while remaining unbiased. Stein’s identity [Stein, 1972] provides a systematic way to construct such zero-mean functions from the score function—i.e., the gradient of the log-density—of the target distribution. Building on this principle, control functionals [Oates et al., 2017] and the zero-variance principle [Assaraf and Caffarel, 1999] have demonstrated the effectiveness of score-based control variates for MC integration, and neural networks have been used to parameterize them in gradient estimation [Grathwohl et al., 2018], rendering [Müller et al., 2020], general MC integration [Wan et al., 2020], and lattice field theory [Bedaque and Oh, 2024, Oh, 2025]. While these methods achieve impressive variance reduction, they all require retraining for each new observation $\mathbf{y}$, precluding amortization—a crucial limitation when retraining for every new observation is computationally prohibitive.

To address this limitation, we introduce conditional neu-

ral control variates (CNCV), a modular variance reduction method for MC estimation of posterior expectations in Bayesian inverse problems. Our method requires: (i) joint samples $(\mathbf{x}, \mathbf{y})$ drawn from the prior predictive distribution—by sampling $\mathbf{x}$ from the prior and computing $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \epsilon$ —samples that are already available in simulation-based inference pipelines [Cranmer et al., 2020]; and (ii) the posterior score function $\nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y})$, which can come from physics-based evaluations, neural operator surrogates [Li et al., 2021], or learned generative models [Rezende and Mohamed, 2015, Song and Ermon, 2019, Ho et al., 2020, Papamakarios et al., 2021, Baldassari et al., 2023]. Once trained, the control variates apply to any new observation $\mathbf{y}$ without retraining, integrating seamlessly with simulation-based inference techniques. In practice, CNCV drops into such a pipeline in three steps: (1) train an amortized posterior—for instance a conditional normalizing flow—on the joint samples; (2) train CNCV on the same samples using that posterior’s score; and (3) at inference, draw samples, evaluate the score, and subtract the learned control variate. While we focus on Bayesian inverse problems, the method applies wherever Monte Carlo estimation is the bottleneck and a smooth conditional score is available—for example, reinforcement-learning policy gradients [Greensmith et al., 2004], latent uncertainty quantification with functional autoencoders [Bunker et al., 2025], and amortized uncertainty quantification for digital twins [Herrmann, 2023].

A further challenge arises when scaling Stein-based control variates to high dimensions: computing the divergence of a general neural network—required by the construction in Section 3—involves $\mathcal{O}(d)$ backward passes or stochastic trace estimation [Hutchinson, 1990]. To address this, we introduce an ensemble of hierarchical affine coupling layers inspired by HINT [Kruse et al., 2021], each operating on a random permutation of the input dimensions. The triangular Jacobian structure enables exact divergence computation in a single forward pass. Each ensemble member captures different cross-dimension interactions through its permutation, and averaging yields control variates with high correlation across all components of the quantity of interest.

Our main contributions are: (i) we extend neural control variates to the conditional setting, yielding amortized control variates that integrate with any posterior sampler and score source; (ii) we introduce an ensemble of hierarchical coupling layers with random input permutations that enable exact divergence computation; and (iii) we validate CNCV on four inverse problems of increasing complexity—Gaussian, Rosenbrock, nonlinear, and PDE-constrained Darcy flow—where it outperforms classical Stein control variates and reduces variance across observations, even under learned scores and non-Gaussian noise.

2 BAYESIAN INVERSE PROBLEMS

We are concerned with estimating an unknown parameter $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^d$ from noisy observations $\mathbf{y} \in \mathcal{Y} \subset \mathbb{R}^m$ related through the forward model $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \epsilon$, where $\mathcal{F} : \mathcal{X} \rightarrow \mathcal{Y}$ is the forward operator and $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ is additive Gaussian noise. In the Bayesian framework [Tarantola, 2005, Kaipio and Somersalo, 2006], the posterior distribution is given by Bayes’ rule:

$$p_{\text{post}}(\mathbf{x}|\mathbf{y}) \propto p_{\text{like}}(\mathbf{y}|\mathbf{x}) p_{\text{prior}}(\mathbf{x}), \quad (1)$$

where $p_{\text{like}}(\mathbf{y}|\mathbf{x})$ is the likelihood, $p_{\text{prior}}(\mathbf{x})$ encodes prior beliefs on the unknown, and the normalizing constant is the marginal likelihood $p(\mathbf{y})$. For brevity, we write $p(\mathbf{x}|\mathbf{y}) \equiv p_{\text{post}}(\mathbf{x}|\mathbf{y})$ throughout. Since the posterior is generally intractable, we must resort to sampling-based methods to extract information from it.

2.1 SAMPLING THE POSTERIOR

Extracting information from the posterior requires access to samples—and many modern posterior sampling methods rely on the *score function* $\nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y})$, either explicitly or implicitly. By Bayes’ rule, the score decomposes as $\nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) = \nabla_{\mathbf{x}} \log p_{\text{like}}(\mathbf{y}|\mathbf{x}) + \nabla_{\mathbf{x}} \log p_{\text{prior}}(\mathbf{x})$. Gradient-based MCMC methods—such as MALA and HMC [Roberts and Rosenthal, 1998, Neal, 2011]—use the score to construct Markov chains that converge to the posterior, but produce correlated samples and require burn-in. Conditional generative models offer an alternative: conditional normalizing flows (CNFs; Rezende and Mohamed, 2015, Papamakarios et al., 2021) learn an invertible map from a base distribution to the posterior, providing both i.i.d. samples and an explicit density from which the score is obtained by automatic differentiation. Diffusion models [Song and Ermon, 2019, Ho et al., 2020, Baldassari et al., 2023] directly parameterize the score via denoising score matching. When the forward operator $\mathcal{F}$ is differentiable—or when its adjoint is available—the likelihood score can also be evaluated analytically [Louboutin et al., 2023]; when $\mathcal{F}$ is expensive, a differentiable surrogate [Li et al., 2021] can approximate it. In all cases, the score of the sampling distribution is available as a byproduct. Our control variate construction (Section 3) exploits this observation, requiring only that the score corresponds to the sampling distribution.

2.2 MONTE CARLO ESTIMATION AND CONTROL VARIATES

Once posterior samples are available, the primary computational task is estimating expectations of quantities of interest $h : \mathcal{X} \rightarrow \mathbb{R}$:

$$\mathbb{E}_{p_{\text{post}}(\mathbf{x}|\mathbf{y})}[h(\mathbf{x})] = \int_{\mathcal{X}} h(\mathbf{x}) p_{\text{post}}(\mathbf{x}|\mathbf{y}) d\mathbf{x}. \quad (2)$$

These expectations provide statistical summaries—e.g., the posterior mean ($h(\mathbf{x}) = \mathbf{x}$), posterior variance ($h(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_{\text{post}})^2$), or predictive quantities. Given N samples $\{\mathbf{x}_i\}_{i=1}^N \sim p(\mathbf{x}|\mathbf{y})$, the MC estimator $\hat{\mathbb{E}}[h(\mathbf{x})] = \frac{1}{N} \sum_{i=1}^N h(\mathbf{x}_i)$ is unbiased with variance $\text{Var}(h(\mathbf{x}))/N$. When $\text{Var}(h(\mathbf{x}))$ is large, accurate estimates demand many samples. A control variate $g: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ satisfying $\mathbb{E}_{p(\mathbf{x}|\mathbf{y})}[g(\mathbf{x}, \mathbf{y})] = 0$ yields the controlled estimator

$$\tilde{\mathbb{E}}[h(\mathbf{x})] = \frac{1}{N} \sum_{i=1}^N (h(\mathbf{x}_i) - g(\mathbf{x}_i, \mathbf{y})), \quad (3)$$

which is also unbiased, with per-sample variance $\text{Var}(h - g) = \text{Var}(h) + \text{Var}(g) - 2\text{Cov}(h, g)$. When g is positively correlated with h and has moderate variance, $\text{Var}(h - g) < \text{Var}(h)$ —i.e., the control variate reduces the per-sample variance. When the samples $\{\mathbf{x}_i\}_{i=1}^N$ are i.i.d. draws from the posterior—e.g., from a conditional generative model or independent importance sampling—the terms $h(\mathbf{x}_i) - g(\mathbf{x}_i, \mathbf{y})$ are themselves i.i.d., and the estimator variance is $\text{Var}(h - g)/N$. The control variate therefore reduces the constant factor from $\text{Var}(h)$ to $\text{Var}(h - g)$ while preserving the $1/N$ convergence rate. The challenge is constructing such zero-mean functions g systematically; our approach (Section 3) uses the score from the sampling distribution to build g via Stein’s identity.

3 CONDITIONAL NEURAL CONTROL VARIATES

The key insight is to combine Stein’s identity—which guarantees zero expectation by construction—with a hierarchical coupling layer architecture that makes the required divergence computation tractable.

3.1 CONTROL VARIATE CONSTRUCTION VIA STEIN’S IDENTITY

Stein’s identity [Stein, 1972] states that for any smooth $\phi: \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}^d$ satisfying appropriate boundary conditions (see Appendix A),

$$\mathbb{E}_{p(\mathbf{x}|\mathbf{y})}[\nabla_{\mathbf{x}} \cdot \phi(\mathbf{x}, \mathbf{y}) + \phi(\mathbf{x}, \mathbf{y}) \cdot \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y})] = 0. \quad (4)$$

The above expression immediately provides a family of control variates parameterized by ϕ :

$$g(\mathbf{x}, \mathbf{y}) = \nabla_{\mathbf{x}} \cdot \phi(\mathbf{x}, \mathbf{y}) + \phi(\mathbf{x}, \mathbf{y}) \cdot \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}), \quad (5)$$

where $\nabla_{\mathbf{x}} \cdot \phi = \sum_{i=1}^d \partial \phi_i / \partial x_i$ denotes the divergence. By construction, $\mathbb{E}_{p(\mathbf{x}|\mathbf{y})}[g(\mathbf{x}, \mathbf{y})] = 0$ for all $\mathbf{y}$, ensuring that g could be used as a control variate.

We parameterize ϕ by a neural network $\phi_{\theta}: \mathbb{R}^{d+m} \rightarrow \mathbb{R}^d$; substituting into equation (5) yields g_{θ} . Unfortunately, computing $\nabla_{\mathbf{x}} \cdot \phi_{\theta} = \sum_{i=1}^d \partial \phi_{\theta,i} / \partial x_i$ for a general neural

network requires d backward passes or stochastic trace estimation [Hutchinson, 1990]—a cost that becomes prohibitive in high dimensions (Appendix I quantifies its accuracy and cost), motivating the architecture below.

3.2 HIERARCHICAL COUPLING LAYER ARCHITECTURE

To address the computational challenge of the divergence, we parameterize ϕ_{θ} using the hierarchical invertible neural transport (HINT) architecture of Kruse et al. [2021], which organizes affine coupling layers [Dinh et al., 2017] in a recursive binary tree. At each node, the input is split into two halves $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\mathbf{x}_1 \in \mathbb{R}^{d_1}$, $\mathbf{x}_2 \in \mathbb{R}^{d_2}$, and transformed via an affine coupling:

$$\phi_{\theta}(\mathbf{x}, \mathbf{y}) = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{s}_{\theta}(\mathbf{x}_1, \mathbf{y}) \odot \mathbf{x}_2 + \mathbf{t}_{\theta}(\mathbf{x}_1, \mathbf{y}) \end{bmatrix} \quad (6)$$

Here, $\mathbf{s}_{\theta}$ and $\mathbf{t}_{\theta}$ are neural networks conditioned on $\mathbf{y}$, and $\odot$ denotes elementwise multiplication. In the recursive tree, the upper half is first processed by its subtree, and the resulting output—rather than the raw $\mathbf{x}_1$ —conditions the coupling of the lower half; the coupled lower half is then processed by its own subtree. This continues until leaf nodes are reached at a prescribed depth. The recursive subdivision ensures that every dimension participates in at least one coupling, producing a lower-triangular Jacobian with all but one diagonal entry learnable.

At a single coupling node, the triangular structure yields diagonal elements $\partial \phi_{\theta,i} / \partial x_i = 1$ for i in the upper partition and $\partial \phi_{\theta,i} / \partial x_i = s_{\theta,i}(\mathbf{x}_1, \mathbf{y})$ for i in the lower partition. The divergence contribution from one node is thus

$$\sum_{i \in \mathbf{x}_2} s_{\theta,i}(\mathbf{x}_1, \mathbf{y}) + d_1, \quad (7)$$

where $d_1 = \dim(\mathbf{x}_1)$. The full-tree Jacobian diagonal is computed recursively: $\mathbf{d} = [\mathbf{d}_{\text{upper}}; \mathbf{s} \odot \mathbf{d}_{\text{lower}}]$, with $\mathbf{d} = \mathbf{1}$ at leaf nodes (see Appendix A). The divergence $\nabla_{\mathbf{x}} \cdot \phi_{\theta} = \mathbf{1}^{\top} \mathbf{d}$ is exact and obtained in a single forward pass. This exactness is decisive at scale: at $d=100$, matching the accuracy of the analytical diagonal with a generic black-box ϕ via Hutchinson trace estimation [Hutchinson, 1990] requires roughly 100 random probes, still incurs an 8–11% relative trace error, and costs about $100\times$ more per evaluation; the triangular Jacobian structure is therefore a prerequisite for tractable, low-variance divergence estimation in high dimensions, not merely a convenience (Appendix I).

Each hierarchical coupling tree ϕ_{θ} maps $\mathbf{x} \in \mathbb{R}^d$ to $\mathbb{R}^d$. Crucially, the full Jacobian diagonal $\text{diag}(\nabla_{\mathbf{x}} \phi_{\theta})$ is available from the recursive forward pass, enabling a *vector-valued* control variate for $\mathbf{h}(\mathbf{x}) \in \mathbb{R}^k$ using a single network:

$$\mathbf{g}_{\theta}(\mathbf{x}, \mathbf{y}) = \text{diag}(\nabla_{\mathbf{x}} \phi_{\theta}) + \phi_{\theta}(\mathbf{x}, \mathbf{y}) \odot \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}), \quad (8)$$

In the above expression, $\odot$ denotes elementwise multiplication and k denotes the dimension of the quantity of interest. Since the j -th component $g_{\theta,j} = \partial \phi_{\theta,j} / \partial x_j + \phi_{\theta,j} \cdot \nabla_{x_j} \log p(\mathbf{x}|\mathbf{y})$ is exactly the j -th term of the product rule expansion *before* summing over components (see Appendix A), each component independently has zero expectation: $\mathbb{E}_{p(\mathbf{x}|\mathbf{y})}[\mathbf{g}_\theta] = \mathbf{0}$. A single tree thus produces control variates for all k components simultaneously, without requiring a separate network per component.

While a single tree produces valid control variates, it is limited by its fixed binary subdivision pattern. To improve coverage, we use L independent ensemble members, each with its own parameters and a random permutation $\mathbf{P}_\ell$ applied to the input dimensions (outputs are mapped back before computing g_j). Since the trace is invariant under conjugation, $\text{tr}(\mathbf{P}_\ell \mathbf{J} \mathbf{P}_\ell^{-1}) = \text{tr}(\mathbf{J})$, each permuted tree still satisfies Stein’s identity (Appendix A), and different permutations capture complementary cross-dimension interactions. The averaged control variate

$$\mathbf{g}_{\text{ens}}(\mathbf{x}, \mathbf{y}) = \frac{1}{L} \sum_{\ell=1}^L \mathbf{g}_{\theta_\ell}(\mathbf{x}, \mathbf{y}) \quad (9)$$

preserves $\mathbb{E}_{p(\mathbf{x}|\mathbf{y})}[\mathbf{g}_{\text{ens}}(\mathbf{x}, \mathbf{y})] = \mathbf{0}$; we verify this empirically in Appendix A (Figure 12). We ablate the effect of ensemble size in Section 4.1.4.

3.3 TRAINING

Our purpose is to find parameters θ that minimize the total variance of the controlled estimator $\mathbf{h} - \mathbf{g}$. Since $\mathbb{E}_{p(\mathbf{x}|\mathbf{y})}[\mathbf{g}_\theta(\mathbf{x}, \mathbf{y})] = \mathbf{0}$ by Stein’s identity, we have

$$\mathbb{E}[\|\mathbf{h} - \mathbf{g}\|^2] = \sum_{j=1}^k \text{Var}(h_j - g_j) + \|\mathbb{E}[\mathbf{h}]\|^2, \quad (10)$$

where all expectations are conditional on the observation $\mathbf{y}$ and the second term is constant in θ . Expanding each component gives $\text{Var}(h_j - g_j) = \text{Var}(h_j) + \text{Var}(g_j) - 2\text{Cov}(h_j, g_j)$, so minimizing the MSE simultaneously penalizes large $\text{Var}(g_j)$ and rewards high $\text{Cov}(h_j, g_j)$ —automatically balancing the two contributions to variance reduction. This yields the training objective:

$$\{\theta_\ell^*\} = \arg \min_{\{\theta_\ell\}} \mathbb{E}[\|\mathbf{h} - \mathbf{g}_{\text{ens}}\|^2], \quad (11)$$

where the expectation is over $(\mathbf{x}, \mathbf{y}) \sim p(\mathbf{x}, \mathbf{y})$. In practice, the expectation is approximated using N training samples $\{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ generated by sampling $\mathbf{x}_i \sim p_{\text{prior}}(\mathbf{x})$ and computing $\mathbf{y}_i = \mathcal{F}(\mathbf{x}_i) + \epsilon_i$. The score function $\nabla_{\mathbf{x}} \log p(\mathbf{x}_i|\mathbf{y}_i)$ is evaluated at each training pair using any of the sources described in Section 2.1.

3.4 INFERENCE

For a new observation $\mathbf{y}_{\text{obs}}$, we draw M posterior samples using any available sampler (cf. line 9 of Algorithm 1) and

Algorithm 1 Conditional neural control variates (CNCV)

Require: Prior p_{prior} , forward model $\mathcal{F}$, score source, posterior sampler, ensemble size L , quantity of interest $\mathbf{h}$

```

1: // Offline training
2: for  $i = 1, \dots, N_{\text{train}}$  do
3:   Sample  $\mathbf{x}_i \sim p_{\text{prior}}(\mathbf{x})$ 
4:   Compute  $\mathbf{y}_i = \mathcal{F}(\mathbf{x}_i) + \epsilon_i$ 
5:   Evaluate  $\nabla_{\mathbf{x}} \log p(\mathbf{x}_i|\mathbf{y}_i)$  from chosen score source
6: end for
7: Initialize  $L$  hierarchical coupling layers with random
   input permutations
8: Optimize  $\{\theta_\ell\}_{\ell=1}^L$  via equation (11)
9: // Online inference (for any new  $\mathbf{y}_{\text{obs}}$ )
10: Draw  $M$  posterior samples  $\{\mathbf{x}_j\}_{j=1}^M \sim p(\mathbf{x}|\mathbf{y}_{\text{obs}})$ 
11: for  $j = 1, \dots, M$  do
12:   Evaluate score  $\nabla_{\mathbf{x}} \log p(\mathbf{x}_j|\mathbf{y}_{\text{obs}})$ 
13:   Evaluate  $\mathbf{g}_{\text{ens}}(\mathbf{x}_j, \mathbf{y}_{\text{obs}})$  via equation (9)
14: end for
15: return  $\hat{\mathbb{E}}[\mathbf{h}] = \frac{1}{M} \sum_{j=1}^M (\mathbf{h}(\mathbf{x}_j) - \mathbf{g}_{\text{ens}}(\mathbf{x}_j, \mathbf{y}_{\text{obs}}))$ 

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compute the variance-reduced estimate (line 15):

$$\hat{\mathbb{E}}[\mathbf{h}(\mathbf{x})] = \frac{1}{M} \sum_{j=1}^M (\mathbf{h}(\mathbf{x}_j) - \mathbf{g}_{\text{ens}}(\mathbf{x}_j, \mathbf{y}_{\text{obs}})). \quad (12)$$

The entire inference procedure involves only neural network forward passes and score evaluations— L coupling layer and score evaluations per sample. When the dominant cost is the forward model—e.g., PDE solves—this overhead is negligible. Crucially, no retraining is required for new observations, maintaining full amortization. Because the offline training cost is fixed while a per-observation neural Stein control variate [Bedaque and Oh, 2024, Oh, 2025] must be refit for every new observation, CNCV amortizes against the strongest such baseline at approximately 190 observations on the Gaussian $d=4$ problem.

4 NUMERICAL EXPERIMENTS

We validate the proposed CNCV framework on four inverse problems of increasing complexity. We first consider three stylized problems—Gaussian, Rosenbrock, and nonlinear—with tractable posterior scores, then scale to a PDE-constrained Darcy flow inverse problem where the score must be learned from data. Throughout, we report the *variance reduction factor* ($\text{VRF} = \text{Var}(h - g) / \text{Var}(h)$), where lower values indicate stronger variance reduction ($\text{VRF} = 0$ is perfect, $\text{VRF} = 1$ is no improvement). In all experiments, $k = d$ —i.e., the quantity of interest has the same dimension as the unknown parameter. Code to partially reproduce these results is available on [GitHub](#).

Table 1: Variance reduction for Gaussian posteriors under per-dimension hyperparameter tuning (recipes in Appendix G). VRF = $\text{Var}(h-g)/\text{Var}(h)$; lower is better. Values are mean $\pm$ std across d components, each averaged over 100 observations.

d	Mean Est. VRF	Var. Est. VRF
2	0.009 ± 0.000	0.006 ± 0.002
4	0.017 ± 0.009	0.008 ± 0.003
8	0.034 ± 0.010	0.011 ± 0.003
16	0.083 ± 0.063	0.120 ± 0.102

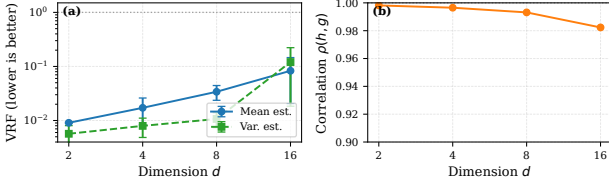


Figure 1: Dimension scaling of CNCV. (a) VRF for mean and variance estimation across $d \in \{2, 4, 8, 16\}$ (lower is better; VRF = 1 means no improvement). (b) Correlation between h and g across dimensions.

4.1 STYLIZED INVERSE PROBLEMS

These experiments quantify variance reduction and its effect on estimation accuracy, study how performance scales with dimension and sample size, ablate the ensemble size, and assess sensitivity to learned scores. Full hyperparameters are in Appendix G.

4.1.1 Gaussian posterior

Consider the inverse problem $\mathbf{y} = \mathbf{x} + \epsilon$ with $\mathbf{x} \in \mathbb{R}^d \sim \mathcal{N}(\mathbf{0}, \Sigma_{\text{prior}})$ and $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ with $\sigma = 0.3$. The prior covariance matrix is a randomly generated positive definite matrix. The analytical posterior is Gaussian with known mean and covariance. We evaluate (i) posterior mean estimation with $\mathbf{h}(\mathbf{x}) = \mathbf{x}$, and (ii) posterior variance estimation with $\mathbf{h}(\mathbf{x}) = (\mathbf{x} - \mu_{\text{post}})^2$.

Table 1 summarizes results across dimensions. We observe that for mean estimation, CNCV achieves VRF between 0.009 and 0.083 across $d \in \{2, 4, 8, 16\}$, while for variance estimation, VRF remains below 0.02 at $d \leq 8$ and reaches 0.12 at $d = 16$ (each entry is averaged over 100 test observations). Figure 1 shows that VRF grows mildly with d , while correlation between h and g remains above 0.98 for mean estimation. Figure 2 shows the sample efficiency for $d = 4$. We observe from panel (a) that the VRF is sample-size invariant at ~ 0.015 across sample sizes from 10 to 5k. Panel (b) shows MSE vs. sample size: while both estimators follow the $1/N$ rate, the CV estimator has a substantially

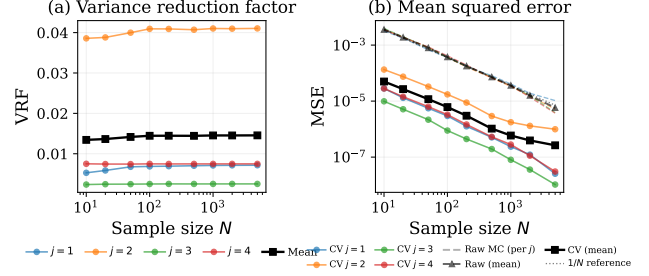


Figure 2: Sample efficiency for $d = 4$ mean estimation. (a) VRF is sample-size invariant at ~ 0.015 . (b) MSE follows $1/N$ rate; the CV estimator provides a $\sim 65\times$ effective sample increase. Index j denotes the component.

lower constant factor, a $\sim 65\times$ effective sample-size increase. On this Gaussian target the posterior score is linear, so a classical polynomial Stein control variate [Mira et al., 2013] is already near-optimal and can match CNCV on a single observation; this is the easy special case. The moment the posterior departs from Gaussianity the optimal control variate becomes nonlinear in the score, and the polynomial baseline degrades sharply: under a heavy-tailed Student- t likelihood it increases variance on 95% of held-out observations while CNCV reduces it on every one (Appendix J). The expressive coupling-layer ensemble is therefore essential precisely on the non-Gaussian posteriors that motivate this work.

4.1.2 Rosenbrock posterior

To assess CNCV on non-Gaussian posteriors, we consider the Rosenbrock distribution—a benchmark with a strongly curved, banana-shaped density, with prior $p(\mathbf{x}) \propto \exp(-a(x_1 - \mu)^2 - b(x_2 - x_1^2)^2)$ where $\mu = 0$, $a = 0.5$, $b = 1$. Combined with Gaussian likelihood $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \mathbf{I})$ with $\sigma = 0.3$, the posterior is non-Gaussian and exhibits strong nonlinear correlations between components.

Figure 3 demonstrates amortization across three test observations from diverse regions: $\mathbf{y}^{(1)}$ from the left tail, $\mathbf{y}^{(2)}$ from the right tail at high x_2 , and $\mathbf{y}^{(3)}$ near the ridge. Each observation (marked by stars) induces a differently positioned banana-shaped posterior. Despite this diversity, the *same trained model* achieves VRF between 0.08 and 0.23 across all cases (panel d). This confirms that the control variate generalizes to any new observation.

Aside from posterior mean estimation, CNCV also reduces variance for posterior variance estimation. Figure 4 uses $\mathbf{h}(\mathbf{x}) = (\mathbf{x} - \mu_{\text{est}})^2$, where μ_{est} and the score are obtained from a pretrained CNF (Section 2.1). The first component (x_1) achieves VRF = 0.08, while the banana direction (x_2)—where the posterior exhibits the strongest nonlinear correlations—is harder (VRF = 0.79), yielding a mean VRF of 0.44 across 10 test observations.

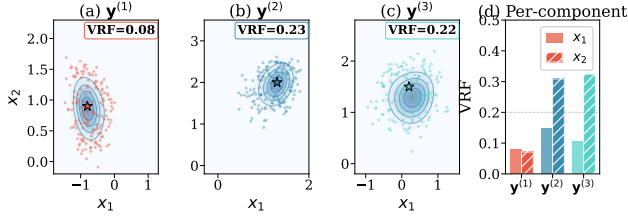


Figure 3: Amortized CNCV on Rosenbrock posterior. (a–c) Test observations from left tail, right tail, and ridge (stars) yield diverse posteriors; VRF values are shown in each panel. (d) Per-component VRF for each observation; the same trained model achieves $\text{VRF} \in [0.08, 0.23]$ across all cases (lower is better).

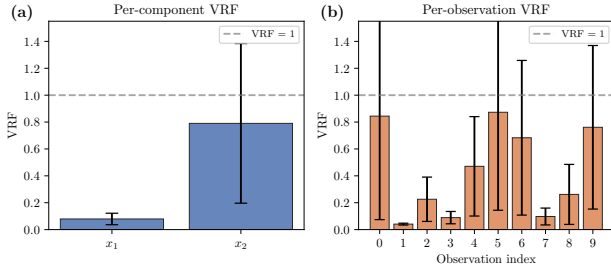


Figure 4: Posterior variance estimation on the Rosenbrock problem. (a) Per-component VRF: x_1 achieves strong reduction while x_2 (banana direction) is harder. (b) Per-observation VRF across 10 test observations (mean 0.44).

4.1.3 Nonlinear forward model

We next consider the nonlinear forward model $\mathcal{F}(\mathbf{x}) = \mathbf{A}\mathbf{x} + \sin(\mathbf{x})$ in $d = 4$, where $\mathbf{A}$ has condition number 2. Figure 5 shows the non-Gaussian posteriors for three test observations. At test time, posterior samples come from a denoising diffusion probabilistic model [Ho et al., 2020] trained on the same joint distribution, providing fast amortized sampling. CNCV achieves $\text{VRF} = 0.57 \pm 0.29$ across 10 test observations. Figure 6 shows the per-component and per-observation VRFs.

4.1.4 Training efficiency and ensemble size

Figure 7(a) shows VRF versus total training samples seen for both the Gaussian ($d = 4$) and Rosenbrock ($d = 2$) problems. We observe that both converge rapidly: the Gaussian VRF drops from ~ 19 to below 0.03, and the Rosenbrock from ~ 277 to below 0.14, within the first few passes over the dataset. Figure 7(b) shows VRF as a function of the ensemble size L . A single member ($L = 1$) yields almost no variance reduction; however, two members already reduce VRF by roughly three orders of magnitude on the Gaussian problem, and performance saturates beyond $L \approx 8$. This motivates the default $L = 16$ used throughout our experi-

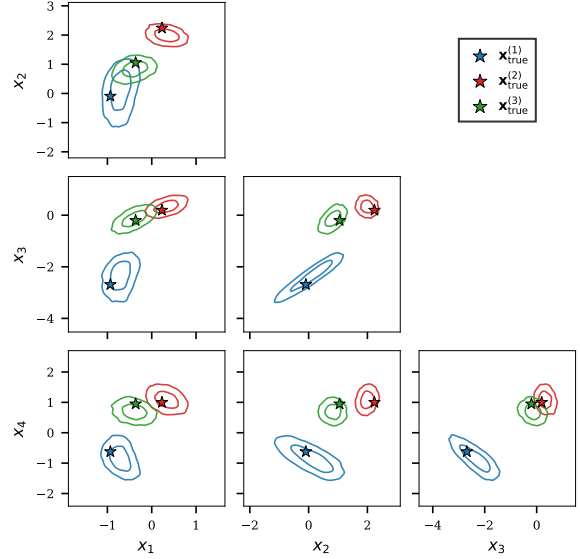


Figure 5: Corner plot of the nonlinear posterior ($d = 4$) for three test observations (blue, red, green). Contours show 50% and 90% HDRs; stars mark the true parameters.

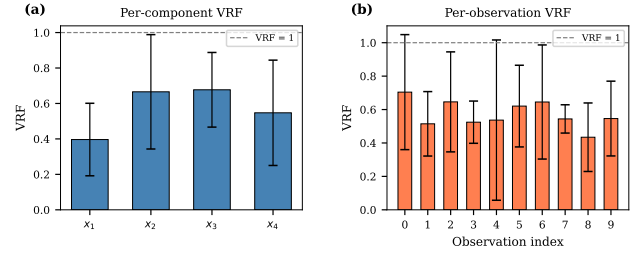


Figure 6: CNCV variance reduction on the nonlinear forward model ($d = 4$). (a) Per-component VRF averaged over 10 test observations. (b) Per-observation VRF (mean 0.57).

ments.

4.1.5 Sensitivity to learned scores

All preceding experiments used posterior scores computed directly from the known likelihood and prior. In practice, however, the score must often be learned from data—particularly for inverse problems governed by expensive PDEs. To assess robustness to score approximation error, we train a CNF to approximate $p(\mathbf{x}|\mathbf{y})$ via negative log-likelihood on joint samples drawn from $p(\mathbf{x}, \mathbf{y})$. The score is then obtained by automatic differentiation of the change-of-variables log-density (CNF architecture details in Appendix G). As discussed in Section 2.1, using the CNF score together with CNF samples yields an estimator unbiased for $\mathbb{E}_{p_\varphi}[h(\mathbf{x})]$ —i.e., expectations under the learned posterior rather than the true posterior. MSE in Figure 8 is accordingly computed against the CNF sample mean. Table 2 compares analytical and CNF-learned scores on all three

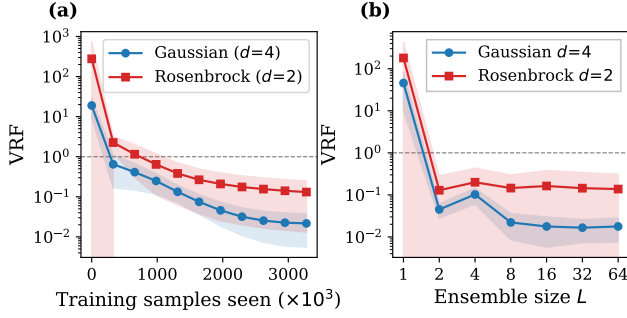


Figure 7: (a) VRF versus total training samples seen. (b) VRF versus ensemble size L . Shading shows ± 1 standard deviation across test observations. Both panels show Gaussian $d=4$ and Rosenbrock $d=2$ mean estimation, using the reference configuration to isolate training dynamics; the per-dimension-tuned results are in Table 1.

Table 2: Analytical vs. CNF-learned score comparison. VRF = $\text{Var}(h-g)/\text{Var}(h)$; lower is better. Gaussian and nonlinear: mean $\pm$ std across components; Rosenbrock: range over three test observations. Each ensemble is trained and evaluated with the same score source. The Gaussian architecture is held fixed at the reference configuration (64 hidden units, 3-layer MLP) to isolate the score-source effect; the per-dimension-tuned results appear in Table 1.

Problem	Score	VRF	Red. (%)	ρ
Gauss. $d=4$	Analyt.	0.040 ± 0.011	96	>0.99
Gauss. $d=4$	CNF	0.058 ± 0.002	94	0.98
Rosenb. $d=2$	Analyt.	0.08–0.23	77–92	—
Rosenb. $d=2$	CNF	0.010 ± 0.002	99	>0.99
Nonlin. $d=4$	Analyt.	0.57 ± 0.29	43	—
Nonlin. $d=4$	CNF	0.57 ± 0.26	43	—

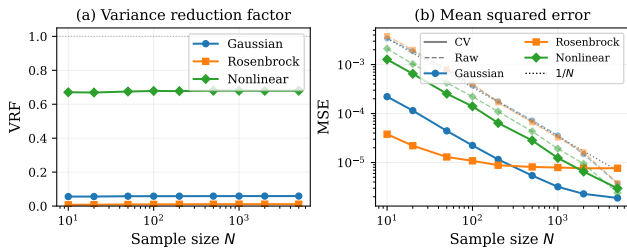


Figure 8: Sample efficiency with CNF-learned scores. (a) VRF is sample-size invariant. (b) The CV estimator (solid) provides a constant-factor MSE improvement over raw MC (dashed). MSE plateaus at large N due to residual bias from the CNF score approximation.

benchmarks. On the Gaussian problem ($d=4$), the CNF score increases VRF from 0.040 to 0.058—a modest degradation consistent with score approximation error. On the Rosenbrock problem ($d=2$), the CNF score achieves VRF of 0.010 ± 0.002 with $\rho > 0.99$, lower than the analytical-

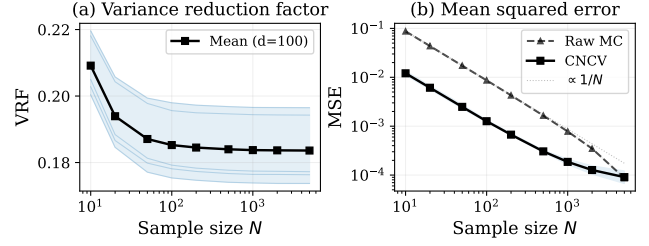


Figure 9: Sample efficiency for Darcy flow. (a) VRF is sample-size invariant at ~ 0.18 . (b) MSE of the posterior mean estimator follows $1/N$ for both raw MC and CNCV, with the CV estimator achieving a $\sim 5.5\times$ constant-factor improvement. At large N , both curves plateau because the reference mean is estimated from a finite CNF sample pool.

score VRF (0.08–0.23)—likely because the CNF provides i.i.d. samples, whereas the analytical-score experiment relies on correlated MCMC draws. On the nonlinear problem ($d=4$), the CNF score achieves VRF = 0.57 ± 0.26 , virtually identical to the analytical score (0.57 ± 0.29), despite 17.5% relative score error. These results suggest that the coupling layer architecture is robust to moderate score approximation errors. Appendix K confirms this directly: with the trained ensemble held fixed and the analytical Gaussian score perturbed by controlled noise, the VRF degrades gracefully and monotonically, rising only to 0.35 even at a 57% relative score error—well beyond the error of the learned scores used here. Figure 8 confirms that VRF remains sample-size invariant with CNF-learned scores, and the CV estimator maintains a constant-factor MSE improvement over raw MC across all sample sizes.

4.2 PDE-CONSTRAINED INVERSE PROBLEM

We now apply CNCV to a higher-dimensional, PDE-constrained problem: estimating the log-permeability field in a groundwater flow model governed by Darcy’s equation [Darcy, 1856], $-\nabla \cdot (e^{u(s)} \nabla p(s)) = 0$ on $\mathcal{D} = [0, 1]^2$, with mixed boundary conditions [Beskos et al., 2017]. The unknown log-permeability $u(s)$ is observed through $m = 33$ pressure sensors with Gaussian noise ($\sigma_y = 0.01$). The PDE forward model costs ~ 0.5 s per solve, the posterior is high-dimensional, and the score must be learned from data. Following Beskos et al. [2017], the prior on u is a Gaussian random field (GRF) represented via a cosine Karhunen–Loève (KL) expansion with truncation $K = 10$, giving $u(s) = \sum_{j=1}^{100} z_j \sqrt{\lambda_j} \psi_j(s)$ where λ_j are the GRF eigenvalues and ψ_j are cosine eigenfunctions. The training data consists of $d = K^2 = 100$ KL coefficients $\mathbf{z} \in \mathbb{R}^{100}$ obtained by expanding GRF samples, paired with the corresponding observations. The forward model reconstructs the field from $\mathbf{z}$, solves the PDE on a 40×40 grid using Devito [Louboutin et al., 2019, Luporini et al., 2020], and extracts pressures at the sensor locations. Since the poste-

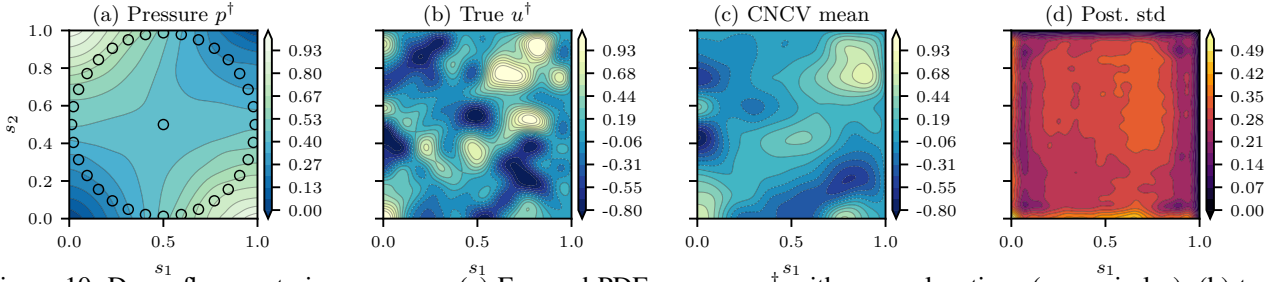


Figure 10: Darcy flow posterior summary. (a) Forward PDE pressure $p^\dagger$ with sensor locations (open circles); (b) true log-permeability $u^\dagger$; (c) CNCV posterior mean from $N = 5,000$ CNF samples; (d) pointwise posterior standard deviation. All fields reconstructed at 128×128 from KL coefficients.

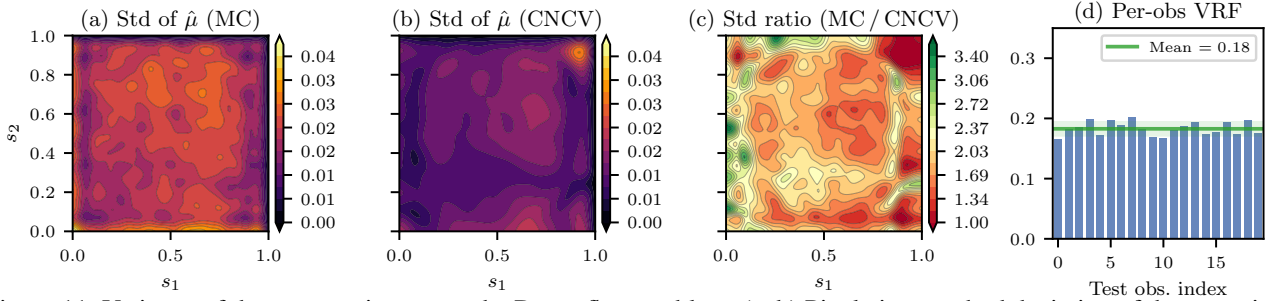


Figure 11: Variance of the mean estimator on the Darcy flow problem. (a, b) Pixelwise standard deviation of the posterior mean estimator $\hat{\mu} = \frac{1}{N} \sum_i \mathbf{x}_i$ for vanilla MC and CNCV; (c) std ratio (MC / CNCV), where values > 1 indicate that CNCV reduces estimator variance; (d) per-observation VRF across 20 held-out test observations.

rior score is not available in closed form for this nonlinear PDE, we train a CNF on 120k joint $(\mathbf{z}, \mathbf{y})$ samples to serve as both posterior sampler and score source (Appendix C verifies the quality of the learned score). The CNCV ensemble uses $L = 16$ depth-3 trees; full hyperparameters are in Appendix G.

Figure 9 confirms that the VRF is sample-size invariant at ~ 0.18 , with the CV estimator achieving $\sim 5.5\times$ lower MSE—an effective $5.5\times$ sample size increase on this 100-dimensional problem. Because CNCV draws its samples from the trained CNF rather than solving the PDE, this gain is effectively free at inference: producing an estimate takes ~ 0.5 s, against ~ 100 s for the 200 PDE solves a raw Monte-Carlo estimate of comparable accuracy would require (Appendix G). Figure 10 shows the posterior summary for a held-out observation: the CNCV mean closely matches the truth (Appendix D quantifies the CNF posterior-mean bias against held-out truth), with posterior uncertainty concentrated in the domain interior—away from sensors and Dirichlet boundaries. Since the reconstruction is linear in the KL coefficients, the posterior standard deviation is governed by the sensor geometry rather than the specific observation values, and is therefore consistent across test cases (Appendix E). As observed from Figure 11, CNCV reduces estimator standard deviation most strongly along large-scale structures; per-observation VRF is tightly concentrated around 0.18 across 20 test observations. To con-

firm that CNCV is not specific to Darcy flow, we apply it to a second, physically distinct PDE-constrained problem—full-waveform inversion governed by the Helmholtz equation [Virieux and Operto, 2009]—where it attains a per-observation VRF of 0.27 ± 0.03 , below unity on all 20 held-out observations (Appendix F).

5 RELATED WORK

Stein-based control variates have a rich history in MC variance reduction. The zero-variance principle of Assaraf and Caffarel [1999] inspired kernel-based control functionals [Oates et al., 2017, 2019, South et al., 2022] and polynomial/regularized MCMC estimators [Mira et al., 2013, South et al., 2023], while Si et al. [2022] unified polynomial, kernel, and neural function classes under a scalable Stein objective. Neural networks have also been used to parameterize control variates outside the Stein framework—for gradient estimation [Grathwohl et al., 2018], rendering [Müller et al., 2020], general MC integration [Wan et al., 2020], and vector-valued [Sun et al., 2023a] and meta-learning [Sun et al., 2023b] settings—and within it, for lattice field theory [Bedaque and Oh, 2024, Oh, 2025, Hasan et al., 2024] and reinforcement-learning policy training [Li et al., 2026]. However, all of these methods are *non-amortized*: each must be fitted for a specific target distribution, requiring MCMC samples and an $\mathcal{O}(n^3)$ kernel solve [Oates et al., 2017],

or least-squares regression on MCMC output [Mira et al., 2013]. In a direct head-to-head on the Gaussian $d=4$ problem, CNCV attains the lowest variance reduction factor among these baselines, and on a bimodal Gaussian mixture it continues to reduce variance as the dimension grows to $d=64$, where the degree-one polynomial basis saturates and the degree-two basis becomes intractable—matching a per-observation neural Stein control variate while training only once (Appendix H).

In contrast, amortized methods train a single model that generalizes across observations [Blei et al., 2017]. Conditional normalizing flows [Rezende and Mohamed, 2015, Papamakarios et al., 2021] and simulation-based inference [Cranmer et al., 2020, Radev et al., 2020] learn density estimators $p_\varphi(\mathbf{x}|\mathbf{y})$ from joint samples, with extensions to inverse problems [Siahkoobi et al., 2022, 2023, Orozco et al., 2025]; importance sampling can further reduce variance through better proposals [Dax et al., 2023], complementary to control variates. However, to our knowledge, no prior work has proposed *amortized control variates*—observation-conditioned functions $g(\mathbf{x}, \mathbf{y})$ with zero posterior expectation that generalize to unseen $\mathbf{y}$ without retraining.

A complementary line of work reduces cost along the axis of forward-model fidelity rather than the observation. Multifidelity and multilevel Monte Carlo methods combine many cheap low-fidelity solves with a few expensive high-fidelity ones to lower the cost of a fixed estimator [Peherstorfer et al., 2018], and multilevel delayed-acceptance MCMC [Lykkegaard et al., 2023] accelerates posterior sampling through a hierarchy of coarsened forward models. These methods operate on an axis orthogonal to CNCV: multifidelity reduces cost over forward-model fidelity, whereas CNCV reduces variance over the observation $\mathbf{y}$. They also differ in their amortization profile—multilevel delayed acceptance is asymptotically exact but non-amortized, running a fresh sampler per observation, while CNCV trains once and generalizes across observations at the cost of inheriting the bias of the learned score. For applications requiring posteriors at more than roughly twenty distinct observations, the amortized cost of CNCV is preferable; for a single high-precision posterior, an asymptotically exact sampler such as multilevel delayed acceptance is the better choice.

6 DISCUSSION

The proposed method integrates seamlessly with simulation-based inference techniques [Cranmer et al., 2020], which already produce joint samples $(\mathbf{x}_i, \mathbf{y}_i)$ and learned posterior approximations from which the score is readily available. CNCV training consumes exactly these existing artifacts, adding no extra forward model evaluations, and the inference-time cost of evaluating the ensemble of coupling layers is comparable to a single shallow network evaluation. At the same time, CNCV also accommodates physics-based

scores computed directly from the likelihood and prior, making it applicable to problems where high-fidelity forward evaluations are available and accurate score computation is paramount. This flexibility makes CNCV attractive across a wide range of settings—from problems constrained by partial differential equations, where each forward solve dominates the computational budget and any variance reduction translates directly into proportional savings, to lower-cost problems where the analytical score is readily accessible.

It is worth noting that CNCV reduces the variance of posterior expectations computed from a given set of samples, but does not alter the samples themselves or improve the underlying posterior approximation. While the quality of the control variate depends on score accuracy, our normalizing flow-based experiments demonstrate that learned scores yield comparable variance reduction to analytical scores. Since Stein’s identity requires the score to correspond to the sampling distribution, it is crucial that the score and posterior samples originate from the same source—e.g., both from a CNF—to preserve the zero-mean property of the control variate. Consequently, CNCV is exactly unbiased for expectations under the learned posterior, and any discrepancy with the true posterior is attributable to the learned score alone—the control variate neither introduces nor amplifies it. On the Darcy problem we quantify this gap directly: the systematic field-space bias of the CNF posterior mean is negligible ($\sim -10^{-3}$), and the residual error is dominated by the irreducible posterior spread on weakly-constrained modes rather than by any bias CNCV contributes (Appendix D). The method also yields little benefit when the quantity of interest is already nearly constant under the posterior—so that its variance is small to begin with—or is only weakly correlated with the score. For Gaussian posteriors, whose score is linear, polynomial control variates [Mira et al., 2013] can match CNCV for a single observation; the value of CNCV lies in non-Gaussian settings, where amortization across observations is needed and the optimal control variate is nonlinear in the score—as Appendix J shows on a heavy-tailed Student- t likelihood, on which a per-observation polynomial control variate instead increases variance. As with any amortized method, we expect a gap relative to observation-specific optimization [Orozco et al., 2025]; CNCV is complementary to such refinement strategies and can be combined with them to further reduce estimator variance.

Beyond the inverse problems studied here, amortized, Stein-based control variates apply to other domains where Monte Carlo estimation is a bottleneck—e.g., variance reduction of policy gradient estimators in reinforcement learning [Green-Smith et al., 2004, Li et al., 2026], scaling to higher-dimensional fields in latent spaces learned by functional autoencoders [Bunker et al., 2025], and amortized uncertainty quantification for digital twins [Herrmann, 2023].

Author Contributions

AS conceived the project, developed the method, implemented the code, designed and ran all experiments, and wrote the paper. HO contributed to the theoretical framework for Stein-based control variates and provided guidance on the neural network parameterization.

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A DERIVATION OF STEIN'S IDENTITY FOR POSTERIOR DISTRIBUTIONS

We provide a self-contained derivation of Stein's identity (equation (4)) in the context of posterior distributions arising from Bayesian inverse problems. The derivation relies on the divergence theorem in $\mathbb{R}^d$ and the product rule for divergences.

A.1 SETUP AND REGULARITY CONDITIONS

Let $p(\mathbf{x}|\mathbf{y})$ denote the posterior density over $\mathbf{x} \in \mathbb{R}^d$ for a fixed observation $\mathbf{y} \in \mathbb{R}^m$, and let $\phi(\mathbf{x}, \mathbf{y}): \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}^d$ be a smooth vector-valued function. We require:

1. $p(\mathbf{x}|\mathbf{y})$ is continuously differentiable and strictly positive on $\mathbb{R}^d$;
2. $\phi(\cdot, \mathbf{y})$ is continuously differentiable for each $\mathbf{y}$;
3. The boundary condition $\lim_{\|\mathbf{x}\| \rightarrow \infty} \phi(\mathbf{x}, \mathbf{y}) p(\mathbf{x}|\mathbf{y}) = \mathbf{0}$ holds, ensuring that the surface integral at infinity vanishes.

The boundary condition is satisfied whenever $p(\mathbf{x}|\mathbf{y})$ decays sufficiently fast—e.g., when the prior is Gaussian or has sub-Gaussian tails—and ϕ grows at most polynomially.

A.2 DERIVATION VIA INTEGRATION BY PARTS

Consider the integral

$$I = \int_{\mathbb{R}^d} \nabla_{\mathbf{x}} \cdot [\phi(\mathbf{x}, \mathbf{y}) p(\mathbf{x}|\mathbf{y})] d\mathbf{x}, \quad (13)$$

where $\nabla_{\mathbf{x}} \cdot \mathbf{v} = \sum_{i=1}^d \partial v_i / \partial x_i$ denotes the divergence. By the divergence theorem applied to the ball $B_R = \{\mathbf{x} : \|\mathbf{x}\| \leq R\}$,

$$\int_{B_R} \nabla_{\mathbf{x}} \cdot [\phi p] d\mathbf{x} = \oint_{\partial B_R} [\phi p] \cdot \hat{\mathbf{n}} dS,$$

where $\hat{\mathbf{n}}$ is the outward normal to the sphere ∂B_R . By the boundary condition (3), the surface integral vanishes as $R \rightarrow \infty$, giving

$$I = \int_{\mathbb{R}^d} \nabla_{\mathbf{x}} \cdot [\phi(\mathbf{x}, \mathbf{y}) p(\mathbf{x}|\mathbf{y})] d\mathbf{x} = 0. \quad (14)$$

A.3 EXPANDING VIA THE PRODUCT RULE

We now expand the divergence in the above expression using the product rule. For each component i :

$$\frac{\partial}{\partial x_i} [\phi_i(\mathbf{x}, \mathbf{y}) p(\mathbf{x}|\mathbf{y})] = \frac{\partial \phi_i}{\partial x_i} p(\mathbf{x}|\mathbf{y}) + \phi_i(\mathbf{x}, \mathbf{y}) \frac{\partial p(\mathbf{x}|\mathbf{y})}{\partial x_i}.$$

Summing over $i = 1, \dots, d$ and using the identity $\nabla p/p = \nabla \log p$:

$$\begin{aligned} \nabla_{\mathbf{x}} \cdot [\phi p] &= \left(\sum_{i=1}^d \frac{\partial \phi_i}{\partial x_i} \right) p + \left(\sum_{i=1}^d \phi_i \frac{\partial p}{\partial x_i} \right) \\ &= (\nabla_{\mathbf{x}} \cdot \phi) p + (\phi \cdot \nabla_{\mathbf{x}} \log p) p. \end{aligned}$$

Substituting into equation (14) and dividing by p inside the expectation:

$$\mathbb{E}_{p(\mathbf{x}|\mathbf{y})} [\nabla_{\mathbf{x}} \cdot \phi(\mathbf{x}, \mathbf{y}) + \phi(\mathbf{x}, \mathbf{y}) \cdot \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y})] = 0, \quad (15)$$

which is exactly equation (4). This establishes that $g(\mathbf{x}, \mathbf{y}) = \nabla_{\mathbf{x}} \cdot \phi + \phi \cdot \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y})$ has zero expectation under the posterior for *any* ϕ satisfying the regularity conditions.

Component-wise identity. Since the divergence theorem applies to each component of ϕp independently, the identity also holds *per component* before summing:

$$\mathbb{E}_{p(\mathbf{x}|\mathbf{y})} \left[\frac{\partial \phi_j}{\partial x_j} + \phi_j \frac{\partial \log p(\mathbf{x}|\mathbf{y})}{\partial x_j} \right] = 0, \quad j = 1, \dots, d.$$

This justifies the vector-valued control variate in equation (8): each component $g_{\theta,j}$ independently has zero expectation, so $\mathbb{E}[\mathbf{g}_{\theta}] = \mathbf{0}$ without requiring the components to sum.

A.4 APPLICATION TO COUPLING LAYER ARCHITECTURES

For our coupling layer parameterization (equation (6)), the divergence takes the explicit form (7). The triangular structure means $\nabla_{\mathbf{x}} \cdot \phi_{\theta}$ involves only the diagonal of the Jacobian, which equals 1 for untransformed components and $s_{\theta,i}(\mathbf{x}_1, \mathbf{y})$ for transformed components. The resulting control variate

$$g_{\theta}(\mathbf{x}, \mathbf{y}) = \underbrace{\sum_{i \in \mathbf{x}_2} s_{\theta,i}(\mathbf{x}_1, \mathbf{y})}_{\text{divergence (exact)}} + \underbrace{d_1 + \phi_{\theta}(\mathbf{x}, \mathbf{y}) \cdot \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y})}_{\text{score term}}$$

In the above expression, $d_1 = \dim(\mathbf{x}_1)$. This control variate satisfies $\mathbb{E}_{p(\mathbf{x}|\mathbf{y})} [g_{\theta}(\mathbf{x}, \mathbf{y})] = 0$ exactly, with no approximation in the divergence computation. This unbiasedness guarantee holds regardless of the neural network parameters θ (provided the boundary conditions are met), and is preserved under ensemble averaging (equation (9)).

For the hierarchical coupling layer, the recursive binary tree structure computes the Jacobian diagonal as $\text{diag}(\nabla_{\mathbf{x}} \phi) = [\mathbf{d}_{\text{upper}}; \mathbf{s} \odot \mathbf{d}_{\text{lower}}]$, where $\mathbf{d}_{\text{upper}}$ and $\mathbf{d}_{\text{lower}}$ are the diagonals from the upper and lower subtrees, and $\mathbf{s}$ is the scale vector from the coupling at the current node. At leaf nodes, $\mathbf{d} = \mathbf{1}$. The divergence is then $\text{tr}(\nabla_{\mathbf{x}} \phi) = \mathbf{1}^{\top} \mathbf{d}$, which remains analytically computable and exact.

A.5 BOUNDARY CONDITIONS FOR POSTERIOR DISTRIBUTIONS IN INVERSE PROBLEMS

We verify that condition (3) holds in typical inverse problem settings. For a Gaussian prior $p_{\text{prior}}(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and Gaussian likelihood $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \mathbf{I})$, the posterior satisfies $p(\mathbf{x}|\mathbf{y}) \leq C \exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_0^{-1}(\mathbf{x} - \boldsymbol{\mu}_0))$ for some $C > 0$, since the likelihood is bounded above by 1. This Gaussian tail decay ensures $\phi(\mathbf{x}, \mathbf{y}) p(\mathbf{x}|\mathbf{y}) \rightarrow \mathbf{0}$ as $\|\mathbf{x}\| \rightarrow \infty$ whenever ϕ grows at most polynomially—which holds for neural networks with bounded (tanh, sigmoid) or polynomially growing (ReLU) activations. For non-Gaussian priors such as the Rosenbrock density $\exp(-a(x_1 - \mu)^2 - b(x_2 - x_1^2)^2)$, the super-exponential decay in all directions similarly ensures the boundary conditions hold.

A.6 EMPIRICAL VERIFICATION

Figure 12 provides an empirical check of Stein’s identity. For each of 250 test observations $\mathbf{y}$, we draw $N = 5,000$ posterior samples and compute the dimension-averaged sample mean of the control variate, $\frac{1}{Nd} \sum_{i,j} g_j(\mathbf{x}_i, \mathbf{y})$, which should be zero by construction. Across all four experiments—Gaussian ($d = 4$, analytical score), Rosenbrock ($d = 2$), nonlinear ($d = 4$), and Darcy ($d = 100$, all three with learned scores)—this quantity is tightly concentrated around zero (mean ≈ 0 , std $\approx 10^{-3}$), confirming that the zero-mean guarantee holds in practice.

B PER-COMPONENT VARIANCE REDUCTION ON DARCY FLOW

Figure 13 shows the per-component VRF for each of the 100 KL coefficients, sorted by decreasing prior eigenvalue. Two regimes are visible:

Low-frequency modes (components 1–30). These modes correspond to large prior eigenvalues λ_j and carry most of the prior variance. Because the data strongly constrains these components, the posterior variance $\text{Var}(z_j|\mathbf{y})$ is already much smaller than the prior variance λ_j^2 . The control variate must match the posterior mean function $\mathbb{E}[z_j|\mathbf{y}]$ —a nonlinear function of the observation—which is harder for the leading modes where the data is most informative. This yields VRF close to 1.

High-frequency modes (components 60–100). These modes have small prior eigenvalues and are barely constrained by the data: $p(z_j|\mathbf{y}) \approx p(z_j) = \mathcal{N}(0, 1)$. Here the posterior score is dominated by the prior score $\nabla_{z_j} \log p(z_j) = -z_j$, which the coupling layers can approximate easily, yielding VRF as low as 0.04.

The component-averaged VRF is approximately 0.16, while

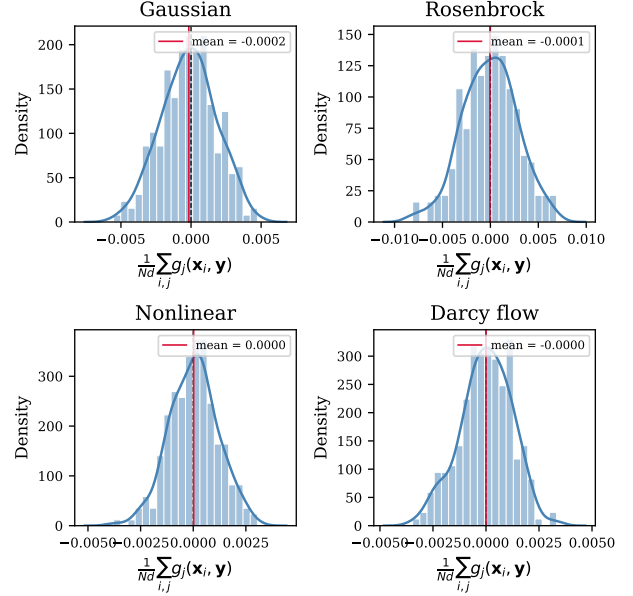


Figure 12: Empirical verification of Stein’s identity. Each panel shows the distribution of $\frac{1}{Nd} \sum_{i,j} g_j(\mathbf{x}_i, \mathbf{y})$ across 250 test observations ($N = 5,000$ samples each). All four distributions are centered at zero, confirming the zero-mean guarantee.

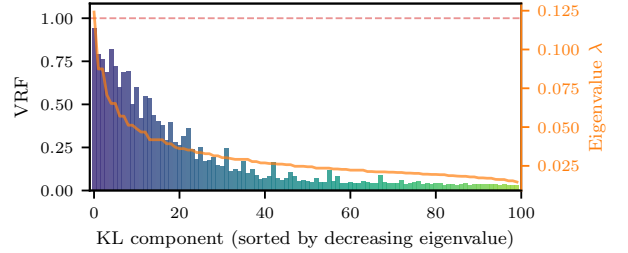


Figure 13: Per-KL-component VRF on the Darcy flow problem ($d = 100$), sorted by decreasing prior eigenvalue λ_j (orange curve, right axis). High-frequency modes (small λ_j , right side) achieve VRF as low as 0.04; low-frequency modes (large λ_j , left side) have VRF ≈ 1 . The dashed line marks VRF = 1 (no improvement).

the per-observation VRF reported in Section 4.2 (~ 0.18) aggregates differently across samples. In both cases, the overall reduction is substantial because the majority of components (those with smaller eigenvalues) achieve very low VRF.

C CNF QUALITY DIAGNOSTICS FOR DARCY FLOW

Since the Darcy flow experiment relies entirely on a learned CNF for both posterior sampling and score computation, we

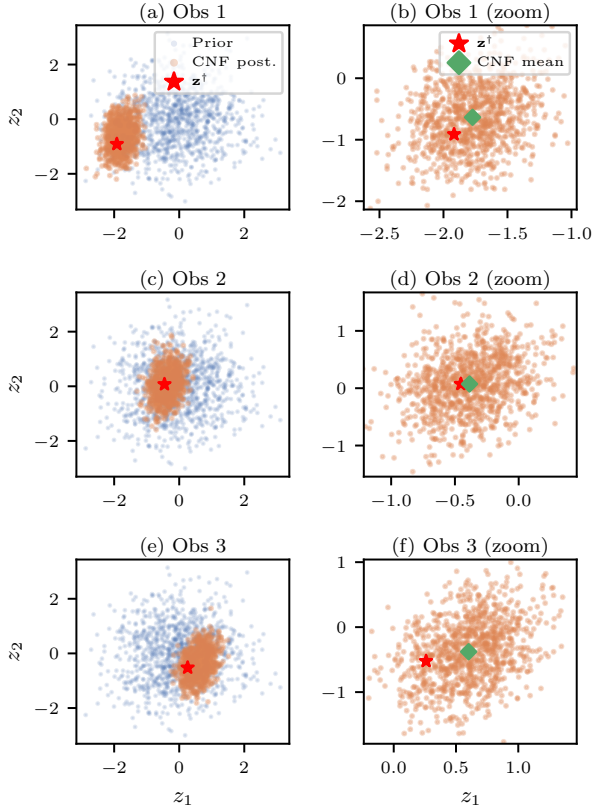


Figure 14: First two KL components of the CNF posterior for three held-out observations. Left column: prior samples (blue) and CNF posterior samples (orange) with the true parameter (red star); the CNF correctly concentrates around the truth. Right column: zoomed view showing the CNF posterior mean (green diamond) near the truth.

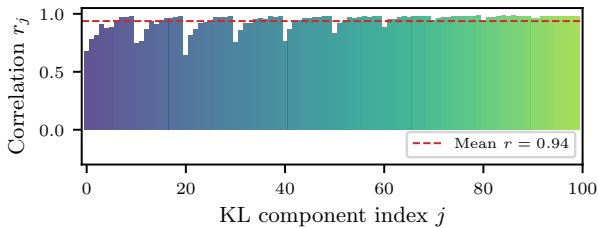


Figure 15: Per-component Pearson correlation between the CNF score $\hat{s}_j = \partial_{z_j} \log p_\varphi(\mathbf{z}|\mathbf{y})$ and the physics-based posterior score s_j (adjoint PDE + prior), evaluated at 100 held-out test observations. Mean $\bar{r} = 0.94$, median 0.97.

provide diagnostics verifying the quality of the trained CNF (Section 4.2).

Figure 14 projects the 100-dimensional posterior onto the first two KL components—i.e., those with the largest prior eigenvalues—for three held-out observations. In each case, the CNF posterior (orange) is correctly concentrated around

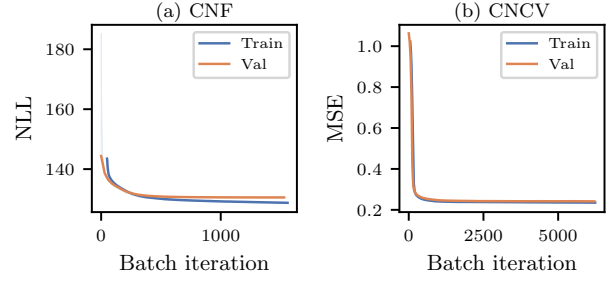


Figure 16: Training and validation loss curves. (a) CNF negative log-likelihood: raw per-batch training loss (light), smoothed training loss (dark), and validation NLL per epoch. (b) CNFV mean-squared error: same layout. Both models converge without overfitting.

the true parameter (red star), with the CNF posterior mean (green diamond) close to the truth. As expected, the prior samples (blue) are substantially more dispersed, confirming that the CNF has learned meaningful posterior concentration.

Figure 15 compares the CNF score $\nabla_{\mathbf{z}} \log p_\varphi(\mathbf{z}|\mathbf{y})$ against the physics-based posterior score computed via adjoint PDE solves and the GRF prior (Section 4.2), for each of the 100 KL components across 100 held-out observations. We observe that the per-component Pearson correlation is high (mean $\bar{r} = 0.94$, median 0.97), confirming that the CNF learns an accurate score across nearly all KL modes. We note that CNFs trained without weight decay ($\lambda = 0$) produced inaccurate scores due to coupling-scale explosion; weight decay of 10^{-4} on the AdamW optimizer was essential for stable training.

Figure 16 shows the training dynamics for both the CNF and the CNFV ensemble. Both models converge smoothly, with validation losses tracking the training losses closely, indicating no significant overfitting.

D CNF POSTERIOR-MEAN BIAS ON DARCY FLOW

The Darcy flow experiment (Section 4.2) relies on a learned CNF for both posterior sampling and the score. As established in Section 2.1, the resulting estimator is unbiased for expectations under the CNF posterior p_φ rather than the true posterior, and the control variate of Section 3 is zero-mean under p_φ for any parameters—so it adds no bias relative to it. The relevant question is then how closely the CNF posterior mean tracks the truth, which we quantify against 20 held-out ground-truth fields.

Figure 17 reports the discrepancy between the CNF posterior mean and the held-out truth, in both the field and KL representations. The field-space bias, averaged over pixels and

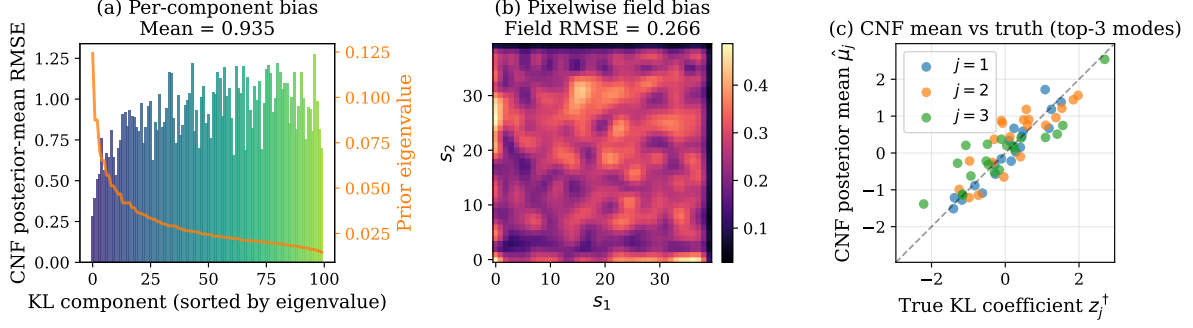


Figure 17: CNF posterior-mean discrepancy against 20 held-out truths on the Darcy problem. The signed field-space bias is negligible ($\sim -10^{-3}$); the unsigned errors—field-space RMSE (~ 0.27) and the per-component KL root-mean-square error, which rises from ~ 0.28 on the most data-informed mode toward ~ 1.0 on weakly-constrained modes—reflect irreducible posterior spread (Appendix B) rather than systematic bias.

observations, is negligible ($\sim -10^{-3}$). The per-component KL root-mean-square error rises from ~ 0.28 on the most data-informed mode (largest prior eigenvalue) toward ~ 1.0 on the weakly-constrained high-frequency modes. As in Appendix B, the latter modes are barely informed by the data, so their posterior approaches the unit-variance prior, and a single posterior-mean estimate necessarily differs from any single truth by an amount of order the prior standard deviation. The aggregate KL error (~ 0.9) is therefore dominated by this irreducible posterior spread rather than by systematic bias. We emphasize that this analysis concerns the CNF posterior relative to the true posterior; quantifying or correcting that gap is distinct from the variance reduction CNCV provides, which is exact with respect to the CNF posterior.

E ADDITIONAL DARCY POSTERIOR VISUALIZATIONS

Figure 18 shows posterior summaries for three additional held-out observations, demonstrating that CNCV generalizes beyond the single observation shown in Figure 10. Each row follows the same format: pressure field $p^\dagger$ with sensor locations, true log-permeability $u^\dagger$, CNCV posterior mean, and posterior standard deviation. Across all observations, the CNCV posterior mean recovers the large-scale structure of the truth, with posterior uncertainty concentrated in the domain interior, away from sensors and Dirichlet boundaries. Figure 19 shows individual posterior draws for the same three observations; the samples retain the high-frequency structure that averaging into the posterior mean smooths out.

F HELMHOLTZ FULL-WAVEFORM INVERSION

To test whether CNCV transfers beyond Darcy flow, we apply it to a second, physically distinct PDE-constrained inverse problem: full-waveform inversion (FWI) governed by the Helmholtz equation, a canonical Bayesian inverse problem in subsurface geophysics [Virieux and Operto, 2009, Beskos et al., 2017]. The unknown is a two-dimensional wave-speed perturbation on a 32×32 grid, parameterized by the same $K = 10$ cosine Karhunen–Loève basis as the Darcy experiment ($d = 100$); the observation $\mathbf{y}$ stacks the complex wavefields recorded at 40 receivers from 5 sources at 4 Hz, with additive noise. As in the Darcy experiment, the posterior score is provided by a conditional normalizing flow trained on joint $(\mathbf{x}, \mathbf{y})$ samples, and CNCV is trained on the same samples using that learned score.

Across 20 held-out observations, CNCV attains a per-observation VRF of 0.27 ± 0.03 , below unity on all 20—a $\sim 3.7\times$ reduction in sample budget at fixed estimator MSE. Figure 20 shows the posterior summary for one observation, Figure 21 the per-component VRF, and Figure 22 the sample efficiency. The empirical Stein-identity check also holds on this second PDE: the dimension-averaged control-variate mean is centered at zero with standard deviation $\sim 10^{-3}$ across 250 test observations, matching the four problems of the main text. CNCV therefore reduces variance on a second, more oscillatory forward operator with no change to the method.

G EXPERIMENTAL DETAILS AND HYPERPARAMETERS

We provide complete hyperparameter specifications for all experiments. All models are trained with the Adam optimizer [Kingma and Ba, 2015] unless otherwise noted (AdamW [Loshchilov and Hutter, 2019] for the Darcy CNF).

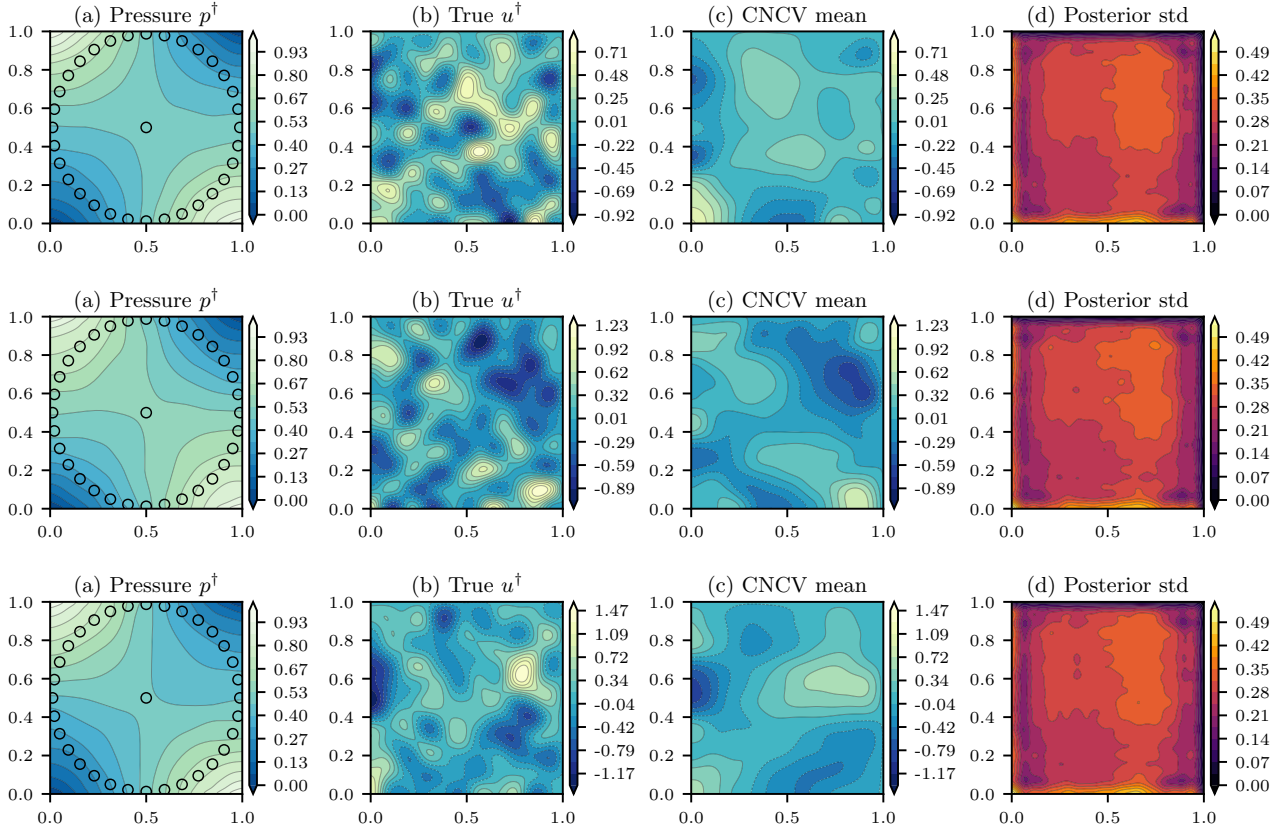


Figure 18: Darcy posterior summaries for three additional held-out test observations. Each row shows a different observation. Columns: (a) pressure $p^\dagger$ with sensors, (b) true $u^\dagger$, (c) CNCV posterior mean, (d) posterior std.

Table 3: CNCV ensemble hyperparameters.

	Gauss.	Rosenb.	Nonlin.	Darcy
d	2–16	2	4	100
L (ensemble)	16	16	16	16
Tree depth	1–3	2	2	3
MLP layers	5	3	3	5
Hidden units	128	64	64	128
Batch size	2048	2048	2048	4096
Train samples	65,536/131,072	65,536	65,536	120,000
Epochs	100	50	100	400
lr_{init}	10^{-4}	10^{-3}	10^{-3}	10^{-4}
lr_{final}	10^{-5}	10^{-4}	10^{-4}	10^{-5}
Seed	12	1	12	12

Learning rates follow a cosine schedule from lr_{init} to lr_{final} .

G.1 CNCV ENSEMBLE (SECTIONS 4.1 AND 4.2)

Table 3 summarizes the CNCV hyperparameters. For the Gaussian benchmark, the configuration is tuned per dimension: the tree depth is 1, 2, 3, 3 for $d = 2, 4, 8, 16$, and the training set comprises 131,072 samples for $d \leq 8$ versus

65,536 for $d = 16$ (the higher-data configuration did not train stably at $d = 16$). The Rosenbrock and nonlinear experiments share a single architecture ($L = 16$, depth 2, 64 hidden, 3-layer MLP), and the Darcy problem requires a deeper MLP (5 layers) and lower learning rate due to the higher dimensionality ($d = 100$).

G.2 CONDITIONAL NORMALIZING FLOWS (SECTIONS 4.1 AND 4.2)

Table 4 shows the CNF hyperparameters. The Gaussian and Rosenbrock CNFs use 12 flow layers with 128 hidden units, while the nonlinear CNF uses 8 flow layers with 64 hidden units; all are trained on 262k joint samples. The Darcy CNF is substantially larger (12 flow layers, 128 hidden, 17.2M parameters) to handle the 100-dimensional KL space with 33-dimensional observations. Weight decay of 10^{-4} is applied only for the Darcy CNF, where it prevents coupling scale explosion (Appendix C).

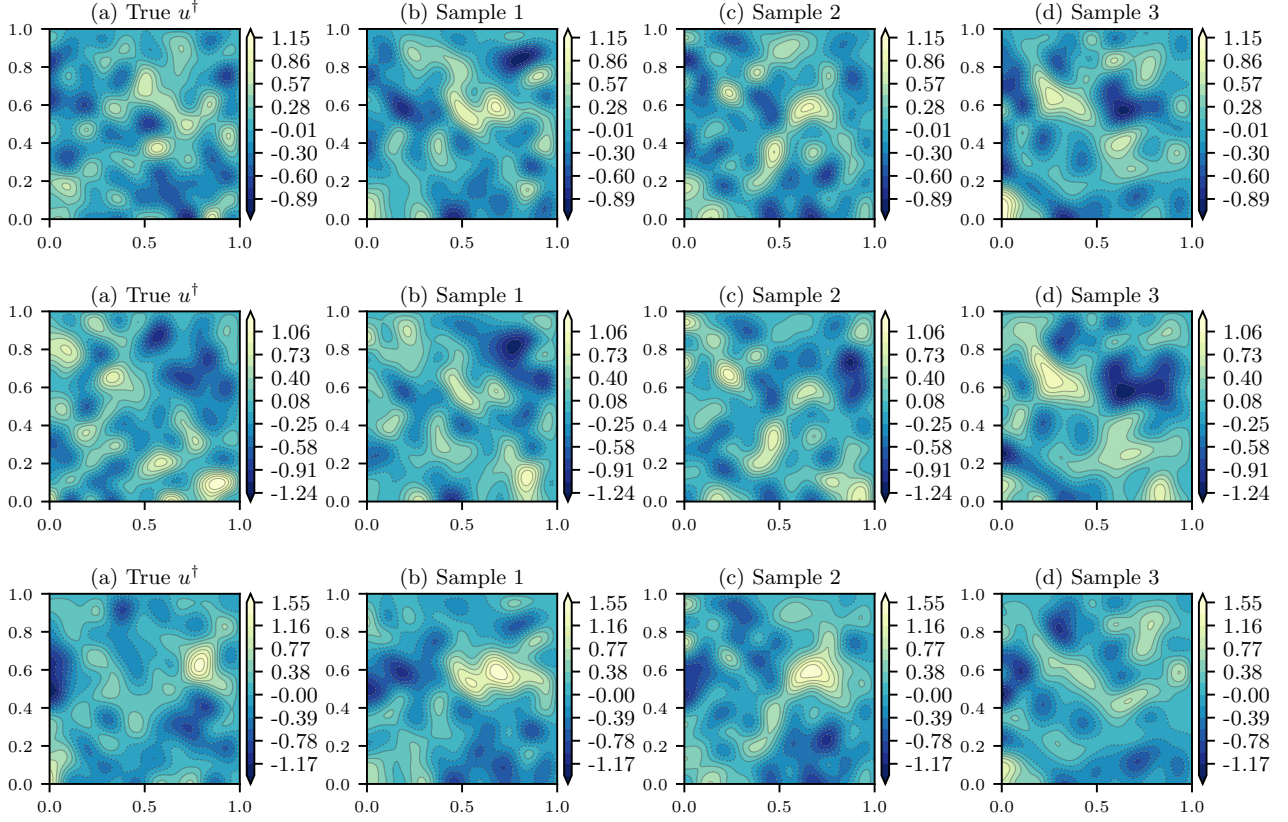


Figure 19: Individual posterior samples for the same three observations as Figure 18. Each row shows the true field (a) and three independent draws from the CNF approximate posterior (b–d). Samples preserve the frequency content of the truth; the smoothness of the posterior mean in Figure 18(c) arises from averaging over uncertain high-frequency modes.

Table 4: Conditional normalizing flow hyperparameters.

	Gauss.	Rosenb.	Nonlin.	Darcy
Flow layers	12	12	8	12
Tree depth	2	2	2	6
MLP layers	3	3	3	3
Hidden units	128	128	64	128
Batch size	8192	4096	8192	4096
Train samples	262,144	262,144	262,144	120,000
Epochs	1000	1000	1000	200
lr_{init}	10^{-3}	10^{-3}	10^{-3}	10^{-3}
lr_{final}	10^{-5}	10^{-5}	10^{-5}	10^{-5}
Weight decay	0	0	0	10^{-4}
Parameters	0.6M	0.2M	0.1M	17.2M

G.3 DIFFUSION MODEL FOR THE NONLINEAR EXPERIMENT

The denoising diffusion probabilistic model used for posterior sampling in the nonlinear experiment (Section 4.1) uses 1000 diffusion timesteps with a linear noise schedule $\beta_t \in [10^{-4}, 0.02]$. The denoising network is a 5-layer MLP with 512 hidden units, conditioned on the observation $\mathbf{y}$ via

concatenation and on the timestep via sinusoidal position embeddings. Training uses 65,536 joint samples, batch size 4096, and a cosine learning rate schedule from 10^{-3} to 10^{-4} over 1000 epochs.

G.4 COMPUTATIONAL COST

Training the CNCF ensemble takes approximately 5 minutes for the stylized experiments ($d \leq 16$) and 50 minutes for the Darcy problem ($d = 100$) on a single NVIDIA RTX 2000 Ada Generation GPU (16 GB). The Darcy CNF requires approximately 2 hours of training. At inference time, evaluating $\mathbf{g}_{\text{ens}}$ for a batch of M samples requires $L = 16$ forward passes through the coupling layer networks plus M score evaluations. For the Darcy problem with $M = 200$ samples, inference takes ~ 0.5 s, compared to ~ 100 s for 200 PDE solves.

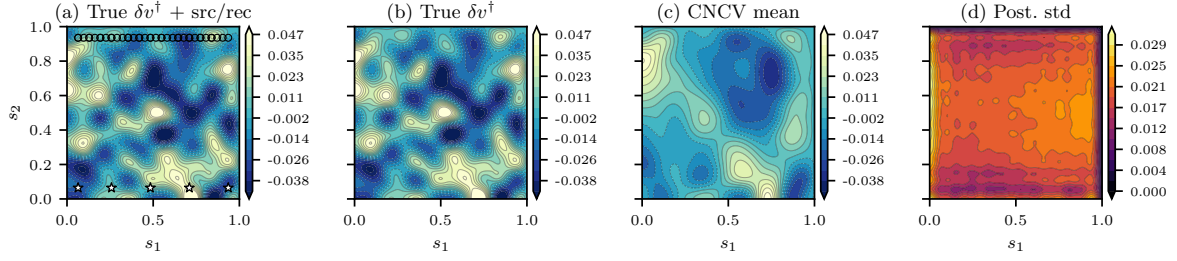


Figure 20: Helmholtz FWI posterior summary for a held-out observation. (a) true wave-speed perturbation $\delta v^\dagger$ with the 5 sources (stars) and 40 receivers (circles); (b) true $\delta v^\dagger$; (c) CNCV posterior mean; (d) posterior standard deviation.

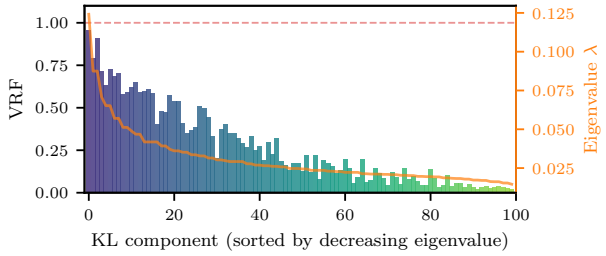


Figure 21: Per-component VRF on the Helmholtz problem ($d = 100$): VRF per KL component (bars), sorted by decreasing prior eigenvalue λ (orange), averaged over 20 held-out observations. The dashed line marks $\text{VRF} = 1$; the data-informed leading modes are hardest, while prior-dominated modes are reduced most.

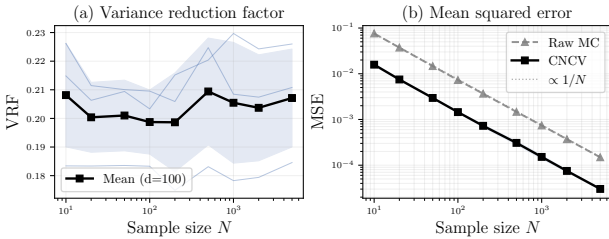


Figure 22: Sample efficiency on the Helmholtz problem. (a) the mean VRF (over the $d = 100$ components) is sample-size invariant; (b) the CNCV estimator (solid) attains a constant-factor reduction in MSE over raw Monte Carlo (dashed), both following the $1/N$ rate.

H COMPARISON WITH CLASSICAL STEIN CONTROL VARIATES

The control variates surveyed in Section 5 are non-amortized and, outside neural parameterizations, restricted to fixed polynomial or kernel function classes. To examine how a low-order polynomial basis behaves as the dimension grows and the posterior becomes multimodal—the regime in which such a basis is least expressive—we compare CNCV against degree-one and degree-two polynomial Stein control

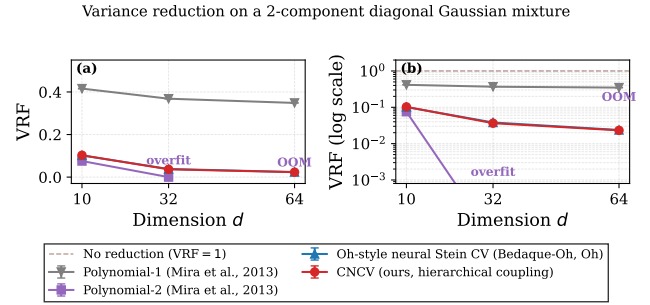


Figure 23: Variance reduction on a bimodal Gaussian mixture versus dimension d . The degree-one polynomial control variate saturates near 0.4 and does not improve with dimension. The degree-two value at $d = 32$ ($\sim 2 \times 10^{-5}$) is an overfit in-sample estimate, fit without a train/test split, and is unavailable at $d = 64$ (out of memory). CNCV reaches ≈ 0.023 at $d = 64$, matching the per-observation neural variant while amortizing across observations.

variates [Mira et al., 2013] and a per-observation neural Stein control variate [Bedaque and Oh, 2024, Oh, 2025].

The target is an equally weighted two-component diagonal Gaussian mixture, with modes separated along every coordinate and per-component scale 0.3, evaluated at $d \in \{10, 32, 64\}$ over three independent repeats. CNCV uses the shared-hierarchical ensemble of Section 3; the polynomial and per-observation neural baselines are fit separately for each observation.

Figure 23 reports the VRF against dimension. The degree-one polynomial control variate saturates near $\text{VRF} \approx 0.4$ and does not improve with dimension, since a linear function of the score cannot capture the bimodal posterior. The degree-two variant is unreliable in this regime: at $d = 32$ its in-sample VRF collapses to $\sim 2 \times 10^{-5}$, but this value reflects overfitting of its $\mathcal{O}(d^2)$ coefficients rather than genuine variance reduction, as it is fit without a train/test split; at $d = 64$ the quadratic feature expansion exhausts memory and no estimate is available. By contrast, both the per-observation neural Stein control variate and CNCV reduce variance increasingly with dimension, reaching $\text{VRF} \approx$

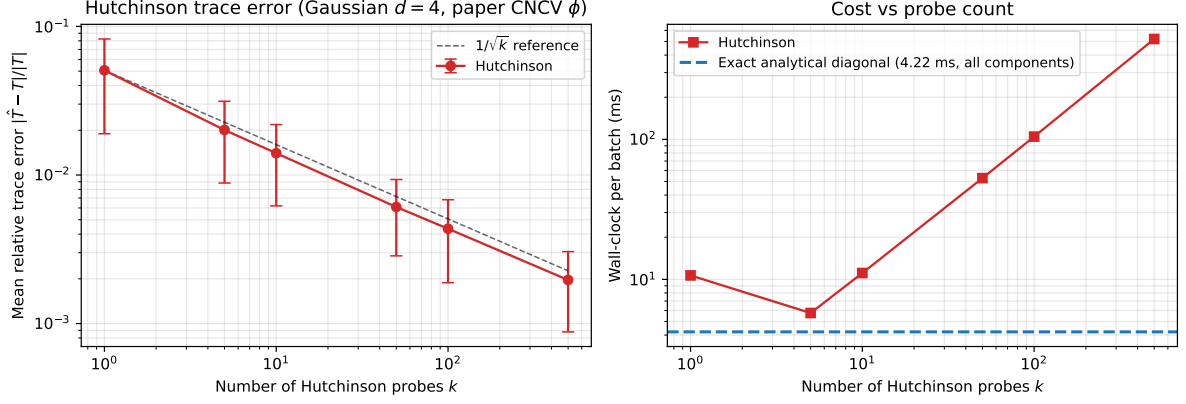


Figure 24: Relative trace error (left axis) and wall-clock time (right axis) of the Hutchinson estimator versus probe count k on the Gaussian problem ($d = 4$), against the exact Jacobian diagonal.

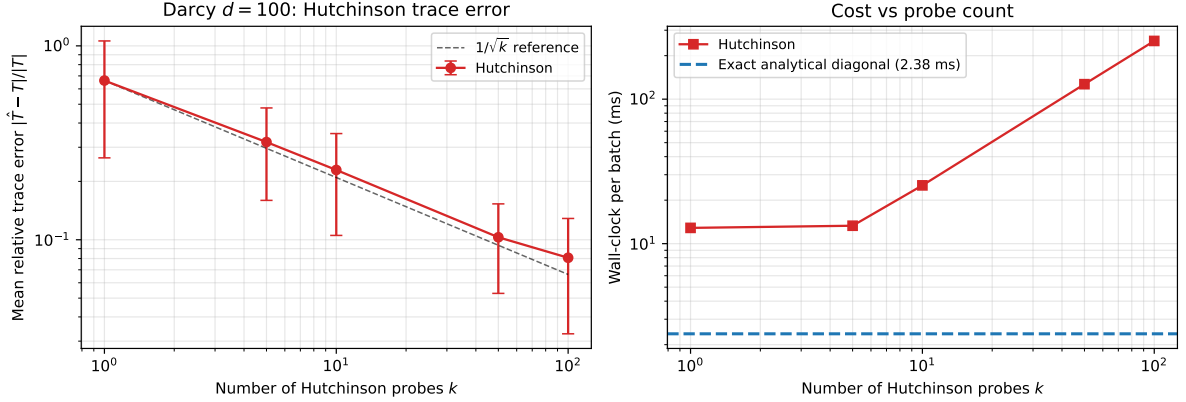


Figure 25: The same comparison on the Darcy problem ($d = 100$); the Hutchinson trace error and its cost relative to the exact diagonal both grow with dimension.

0.023 at $d = 64$ with negligible variation across repeats. CNCV matches the per-observation neural variant (neural 0.024 versus CNCV 0.023 at $d = 64$) while requiring a single training run that generalizes across observations, whereas the neural baseline is refit for every observation.

I EXACT DIVERGENCE VERSUS STOCHASTIC TRACE ESTIMATION

The hierarchical coupling architecture of Section 3 returns the divergence $\nabla_{\mathbf{x}} \cdot \phi$ exactly, as the sum of the Jacobian diagonal computed in a single forward pass. The generic alternative for a black-box ϕ is the Hutchinson estimator [Hutchinson, 1990], which approximates the trace with k random probe vectors. We quantify the accuracy and cost of this alternative against the exact diagonal.

Figures 24 and 25 report the relative trace error and wall-clock time as a function of the probe count k . On the Gaussian problem ($d = 4$), the exact diagonal is computed in ~ 4 ms; matching its accuracy with the Hutchinson estima-

tor requires $k \approx 100$ probes at ~ 105 ms—roughly $25\times$ the cost—and still leaves a relative error of $\sim 0.4\%$. The gap widens sharply with dimension. On the Darcy problem ($d = 100$), a single probe incurs a 66% relative error, and $k = 100$ probes still leave an 8% error at $\sim 106\times$ the cost of the exact diagonal. A second $d = 100$ PDE-constrained problem—full waveform inversion governed by the Helmholtz equation, with the same KL parameterization and a learned CNF score—shows the same pattern: $k = 100$ probes leave an 11% error at $\sim 109\times$ the cost. This widening is expected, since the variance of the Hutchinson estimator scales with the squared Frobenius norm of the Jacobian, which grows with dimension. The exact diagonal is therefore not merely a convenience but a prerequisite for tractable, low-variance divergence estimation at these scales.

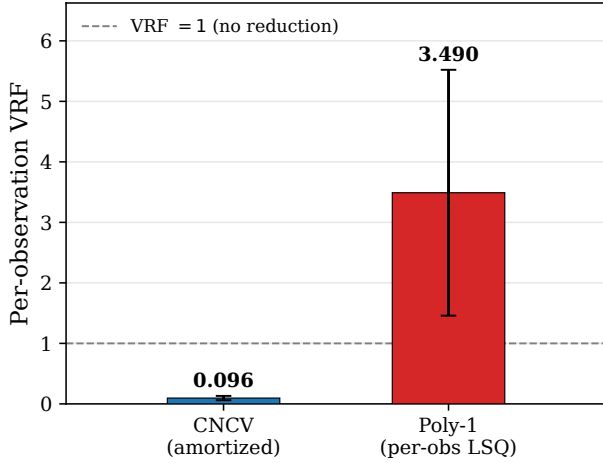


Figure 26: Per-observation VRF under a Student- t ($\nu = 5$) likelihood ($d = 4$, 20 held-out observations). CNCV reduces variance on every observation (mean 0.10), whereas the per-observation linear polynomial Stein control variate increases variance on 95% of them (mean 3.5). The optimal control variate is nonlinear in the saturating Student- t score.

J NON-GAUSSIAN OBSERVATION NOISE

All experiments in Section 4 use Gaussian observation noise. Because the control variate is built from the posterior score rather than any Gaussian assumption (Section 3), it applies unchanged to non-Gaussian likelihoods. We demonstrate this on a heavy-tailed Student- t likelihood.

We take a standard Gaussian prior in $d = 4$ and a component-independent Student- t likelihood with $\nu = 5$ degrees of freedom and scale 0.3, for which the posterior score is available in closed form. For each of 20 held-out observations we draw posterior samples by Langevin dynamics and evaluate the per-observation VRF, comparing CNCV against a per-observation degree-one polynomial Stein control variate fit by least squares.

Figure 26 shows the per-observation results. CNCV reduces variance on every observation (mean VRF 0.10 ± 0.04), whereas the degree-one polynomial control variate increases variance on all but one (mean 3.5): the Student- t score saturates for large residuals, so the optimal control variate is strongly nonlinear—beyond the reach of a linear basis but captured here by the coupling-layer ensemble, the same mechanism that favors CNCV on the non-Gaussian posteriors of Section 4.1.

K ROBUSTNESS TO SCORE APPROXIMATION ERROR

Section 4.1 observed that a learned CNF score yields variance reduction comparable to the analytical score. To probe

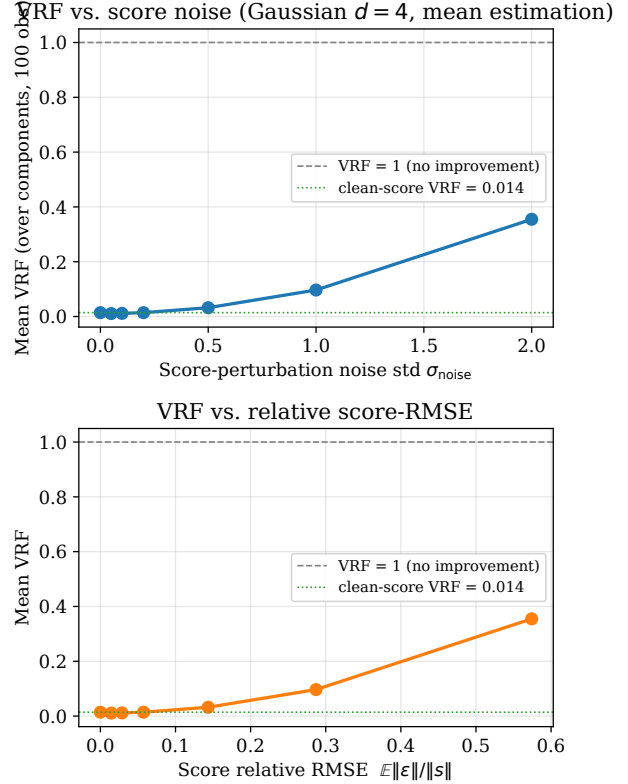


Figure 27: VRF versus relative score error on the Gaussian problem ($d = 4$), perturbing the analytical score with isotropic Gaussian noise while holding the CNCV ensemble fixed. The VRF degrades gracefully: a 14% error yields 0.032, and a 57% error still yields 0.35.

this robustness directly and in isolation, we perturb the analytical Gaussian score ($d = 4$) with isotropic Gaussian noise of increasing magnitude, hold the trained CNCV ensemble fixed, and measure the resulting VRF over 100 observations.

Figure 27 shows that the VRF degrades gracefully and monotonically with the relative score error. A 14% score error raises the VRF only from 0.014 to 0.032, and even a 57% error leaves it at 0.35—still a substantial reduction, consistent with the learned-score result of Section 4.1. This isotropic perturbation does not reproduce the structured error of a learned score, but it isolates the control variate’s sensitivity to score inaccuracy.